

Name : Mr. VIPAN KUMAR GARG
Lab No. : 497244871
Ref By : DR PANKAJ MALHOTRA
Collected : 13/4/2026 9:03:00AM
A/c Status : P
Collected at : FPSC-Sector 8 Panchkula

Age : 69 Years
Gender : Male
Reported : 15/4/2026 3:11:59PM
Report Status : Final
Processed at : LPL-CHANDIGARH LAB
Plot No. 17, Ground Floor, Industrial Area,
Phase-1, Backside Citroen Showroom,
Chandigarh-160002.



Test Report

Test Name	Results	Units	Bio. Ref. Interval
COMPLETE BLOOD COUNT (CBC), WHOLE BLOOD			
Hemoglobin (Photometry)	12.40	g/dL	13.00 - 17.00
Packed Cell Volume (PCV) (Calculated)	36.70	%	40.00 - 50.00
RBC Count (Electrical Impedence)	3.80	mill/mm3	4.50 - 5.50
MCV (Electrical impedence)	96.60	fL	83.00 - 101.00
Mentzer Index (Calculated)	25.4		
MCH (Calculated)	32.80	pg	27.00 - 32.00
MCHC (Calculated)	33.90	g/dL	31.50 - 34.50
Red Cell Distribution Width (RDW) (Electrical Impedence)	16.00	%	11.60 - 14.00
Total Leukocyte Count (TLC) (Electrical Impedence)	3.60	thou/mm3	4.00 - 10.00
Differential Leucocyte Count (DLC)			
Segmented Neutrophils (VCS Technology)	40.40	%	40.00 - 80.00
Lymphocytes (VCS Technology)	36.10	%	20.00 - 40.00
Monocytes (VCS Technology)	14.60	%	2.00 - 10.00
Eosinophils (VCS Technology)	8.40	%	1.00 - 6.00
Basophils (VCS Technology)	0.50	%	<2.00
Absolute Leucocyte Count			
Neutrophils (Calculated)	1.45	thou/mm3	2.00 - 7.00
Lymphocytes (Calculated)	1.30	thou/mm3	1.00 - 3.00
Monocytes (Calculated)	0.53	thou/mm3	0.20 - 1.00
Eosinophils (Calculated)	0.30	thou/mm3	0.02 - 0.50



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Test Name	Results	Units	Bio. Ref. Interval
Basophils (Calculated)	0.02	thou/mm3	0.02 - 0.10
Platelet Count (Electrical impedance)	110	thou/mm3	150.00 - 410.00
Mean Platelet Volume (Coulter Principle)	12.7	fL	6.5 - 12.0
There is leucopenia. Platelets are mildly decreased. Giant platelets seen Platelets are checked manually Advised: Urgent review with Fresh EDTA sample if not correlating clinically.			

Comment

In anaemic conditions Mentzer index is used to differentiate Iron Deficiency Anaemia from Beta- Thalassemia trait. If Mentzer Index value is >13, there is probability of Iron Deficiency Anaemia. A value <13 indicates likelihood of Beta- Thalassemia trait and Hb HPLC is advised to rule out the Thalassemia trait.

Note

- As per the recommendation of International council for Standardization in Hematology, the differential leucocyte counts are additionally being reported as absolute numbers of each cell in per unit volume of blood
- Test conducted on EDTA whole blood



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Test Name	Results	Units	Bio. Ref. Interval
KIDNEY FUNCTION TEST (KFT), SERUM			
Creatinine (Modified Jaffe, IDMS traceable)	1.81	mg/dL	0.70 - 1.30
GFR Estimated (CKD EPI Equation 2021)	40	mL/min/1.73m2	>59
GFR Category (KDIGO Guideline 2012)	G3b		
Urea (Urease, UV)	47.00	mg/dL	17.00 - 49.00
Urea Nitrogen Blood (Urease, UV)	21.95	mg/dL	8.00 - 23.00
BUN/Creatinine Ratio (Calculated)	12		
Uric Acid (Uricase)	6.20	mg/dL	3.50 - 7.20
Total Protein (Biuret)	6.90	g/dL	5.70 - 8.20
Albumin (BCG)	4.60	g/dL	3.20 - 4.60
Globulin(Calculated)	2.30	gm/dL	2.0 - 3.5
A : G Ratio (Calculated)	2.00		0.90 - 2.00
Calcium, Total (Arsenazo III)	10.40	mg/dL	8.80 - 10.20
Result Rechecked, Please Correlate Clinically.			
Phosphorus (Phosphomolybdate, UV)	4.20	mg/dL	2.30 - 3.70
Sodium (Indirect ISE)	141.00	mEq/L	136.00 - 145.00
Potassium (Indirect ISE)	5.54	mEq/L	3.50 - 5.10
Result Rechecked, Please Correlate Clinically.			
Chloride (Indirect ISE)	106.00	mEq/L	98.00 - 107.00

Note



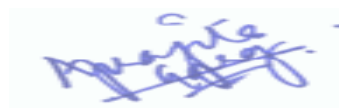
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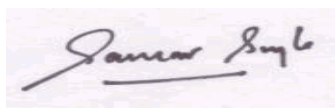


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Test Name	Results	Units	Bio. Ref. Interval
1. Estimated GFR (eGFR) calculated using the 2021 CKD-EPI creatinine equation and GFR Category reported as per KDIGO guideline 2012.			
2. eGFR category G1 or G2 does not fulfil the criteria for CKD, in the absence of evidence of kidney damage			
3. The BUN-to-creatinine ratio is used to differentiate prerenal and postrenal azotemia from renal azotemia. Because of considerable variability, it should be used only as a rough guide. Normally, the BUN/creatinine ratio is about 10:1			
4. Test conducted in Serum			



Dr Aprajita
MBBS,MD(Pathology)
Consultant Pathologist
Dr Lal PathLabs Ltd



Dr. Gaurav Singla
MBBS MD,Pathology
Consultant Pathologist



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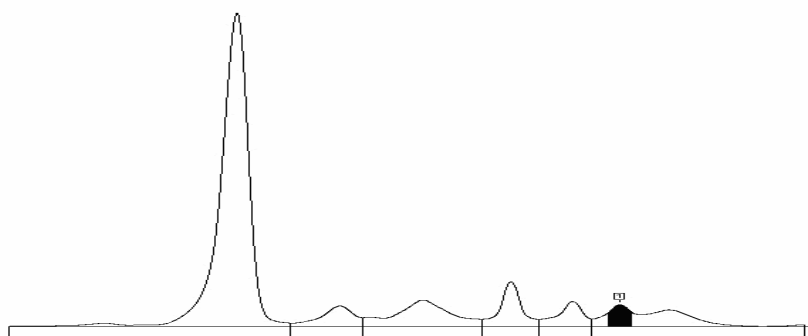
MULTIPLE MYELOMA SCREENING PANEL

PROTEIN ELECTROPHORESIS, SERUM

(Capillary Electrophoresis)

Protein, Total	7.00	g/dL	6.40 - 8.10
Albumin	4.46	g/dL	3.60 - 5.40
Alpha 1 globulin	0.34	g/dL	0.20 - 0.40
Alpha 2 globulin	0.72	g/dL	0.50 - 1.00
Beta 1 globulin	0.47	g/dL	0.50 - 1.10
Beta 2 globulin	0.32	g/dL	0.30 - 0.60
Gamma globulin	0.69	g/dL	0.70 - 1.50
A : G Ratio	1.76		0.90 - 2.00
M Spike	? 0.2	g/dL	

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Interpretation

Suspicious non-discrete protein band in gamma globulin region, possibly Monoclonal gammopathy. Advised: Immunofixation electrophoresis (IFE) to confirm the presence of monoclonal gammopathy and identify its immunological nature.

Comments

Serum Protein electrophoresis (SPE) is used to identify patients with Monoclonal gammopathies (Multiple Myeloma (MM), Waldenstrom's macroglobulinemia, AL Amyloidosis as well as premalignant conditions Monoclonal gammopathy of unknown significance (MGUS) and Smoldering Myeloma) and other serum protein abnormalities (Nephrotic syndrome, Alpha 1 antitrypsin disorder, and inflammatory processes associated with infection, liver diseases & autoimmune disorder). Monoclonal gammopathies indicate clonal expansion of plasma cells or mature B cells. Monoclonal gammopathies are present in 8% of geriatric patients and require further evaluation for monitoring & prognosis. SPE can be used for monitoring response to therapy, a decrease or increase of M spike by 0.5 g/dl is considered significant change.

If SPE alone is used as initial diagnostic screen it will be able to detect approximately 88% of all MM, 66% of AL Amyloidosis and 56% of Light Chain Deposition Disease (LCDD) patients. SPE has good sensitivity in detecting intact monoclonal protein but has limited sensitivity in detecting monoclonal free light chains (FLC). Therefore, SPE alone will miss many patients that are only producing monoclonal FLC. These misses include many Light Chain MM, all Non Secretory MM and significant numbers of AL amyloidosis and LCDD patients. Combination of FLC assay with SPE & immunofixation improves the sensitivity to 99%. Thus, Multiple Myeloma screening panel which includes SPE, IFE and FLC should be done to screen patients with high clinical suspicion for MM.

IMMUNOTYPING, SERUM

(Agarose Gel Electrophoresis and
Immunofixation/ Capillary Electrophoresis and
Immunotyping)

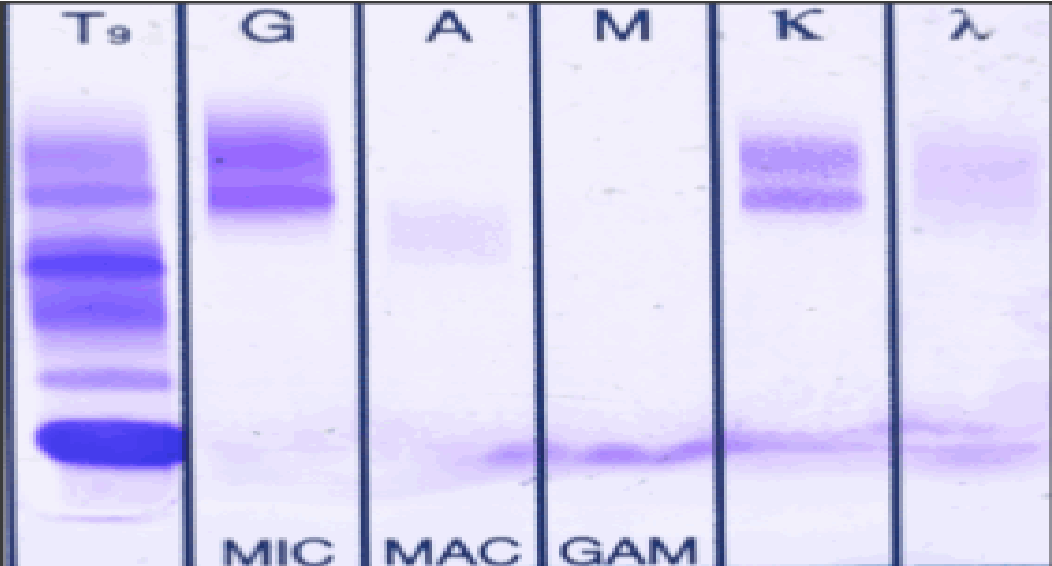


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Test Name	Results			Units	Bio. Ref. Interval	
	T ₉	G	A	M	K	λ
						
		MIC	MAC	GAM		

Interpretation:-Serum protein profile by high-resolution Agarose gel electrophoresis reveals the presence of adiscrete protein band in gamma globulin region suggestive of 'M' spike

A 2nd protein band distal to it is also seen in gamma globulin region could be isomer of the same.

Immunofixation electrophoresis (IFE) identifies the 'M' spike as IgG Kappa and its (?) isomer as IgG Kappa .

Note: Immunotyping is not a quantitative assay. If a monoclonal protein is identified, a serum protein electrophoresis assay is required for quantifying the abnormality.

Comment

The assay is used for immunotyping of monoclonal proteins which identifies the monoclonal immunoglobulin heavy-chain (gamma, alpha, mu) and/or light-chain type (kappa or lambda). It is generally recommended that both serum Protein electrophoresis (SPEP) and Immunotyping be used as a screening panel because it is more sensitive than SPEP. Immunotyping is not only recommended as part of the initial screening process but also for confirmation of complete response to therapy.

Usage

Identification of monoclonal immunoglobulin heavy and light chains
Documentation of complete response to therapy



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Test Name	Results	Units	Bio. Ref. Interval
IMMUNOGLOBULIN PROFILE, SERUM			
Immunoglobulin IgG, Serum (Immunoturbidimetry)	752.00	mg/dL	650.00 - 1600.00
Immunoglobulin IgM, Serum	<21.0	mg/dL	50.00 - 300.00
Immunoglobulin IgA, Serum	88.40	mg/dL	40.00 - 350.00

Comments:

Approximately 80% of serum immunoglobulin is IgG, 6% is IgM and 13% is IgA. High levels of IgM signify acute infections whereas IgG predominates in chronic infections. IgA is the predominant immunoglobulin in body secretions like saliva, sweat and colostrums.

Polyclonal increases are seen in:

- IgG: SLE, Chronic liver diseases, Infectious diseases and Cystic fibrosis
- IgM: Viral, bacterial and parasitic infections, Liver diseases, Rheumatoid arthritis, Scleroderma, Cystic fibrosis & heroin addiction
- IgA: Chronic liver diseases, Chronic infections, Autoimmune disorders, Sarcoidosis and Wiscott-Aldrich syndrome.

Monoclonal increases are seen in IgG & IgA Myelomas and IgM increases in cases of Waldenstroms macroglobulinemia

Decreased synthesis of IgG, IgM & IgA is seen in Congenital and Acquired Immunodeficiency diseases.

Decreased levels are seen in Protein losing enteropathies, Nephrotic syndrome and skin burns.



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Test Name	Results	Units	Bio. Ref. Interval
KAPPA / LAMBDA LIGHT CHAINS, FREE, SERUM (Immunoturbidimetry)			
Kappa, Free Light Chain	97.65	mg/L	3.30 - 19.40
Lambda, Free Light Chain	22.36	mg/L	5.71 - 26.30
Kappa/Lambda(FLC), Ratio	4.367		0.26 - 1.65

Interpretation

KAPPA	LAMBDA	KAPPA / LAMBDA RATIO	REMARKS
Normal	Normal	Normal	Normal sample. If serum electrophoretic pattern is normal, it is unlikely that patient has Monoclonal gammopathy.
Normal	Normal	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
Low	Low	Normal	Suggestive of Bone Marrow suppression
Low	Low	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
High	High	Normal	Suggestive of : • Renal Impairment • Overproduction of Polyclonal free light chain from inflammatory conditions • Biclinal gammopathy of different free light chain types
High	High	Low/High	Suggestive of Monoclonal gammopathy with Renal Impairment

Note: International Myeloma Working Group (IMWG-2014) has updated the criteria for the diagnosis of Multiple Myeloma (MM) which includes clonal bone marrow plasma cells ($\geq 10\%$) or biopsy proven Plasmacytoma with one or more 'Myeloma defining event'. An elevated involved : uninvolved serum FLC ratio >100 (and involved FLC $\geq 100\text{mg/L}$ is now considered a 'Myeloma defining event'.

Comment

Measurement of free light chain (FLCs) aids in the diagnosis & monitoring of Multiple myeloma, Waldenstrom's macroglobulinemia, AL Amyloidosis & Light chain deposition disease. Serum concentration of FLCs is dependent upon the balance between production by plasma cells & their progenitors and renal clearance. When there is increased polyclonal immunoglobulin production and/or renal impairment, concentration of both kappa & lambda light chains can increase by 30-40 fold but the Kappa / Lambda ratio remains unchanged. If



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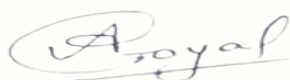
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contrast, tumor produces monoclonal excess of only 1 type of light chains, often with bone marrow suppression of alternate light chains resulting in abnormal kappa/lambda ratio.			

The combination of Serum protein electrophoresis (SPE) & Immunofixation (IFE) are highly sensitive but fail to identify a few patients with Light Chain Multiple Myeloma & all patients of Non Secretory Multiple Myeloma. Combination of free light chain assay with SPE & IFE improves the sensitivity to 99%.

Uses

As per recommendations by IMWG:

- FLC,SPE and serum IFE for screening
- For diagnosis and prognosis of all patients with MGUS, Smoldering or active multiple myeloma, Solitary plasmacytoma and AL amyloidosis
- Serial measurements routinely in patients with AL amyloidosis and Multiple myeloma
- patients with oligosecretory disease
- In all patients who have achieved a complete response (CR) to assess for stringent CR



Dr Anjalika Goyal
MD,Biochemistry (DMC - 67327)
Consultant Biochemist
NRL - Dr Lal PathLabs Ltd



Dr Himangshu Mazumdar
MD, Biochemistry (DMC - 89819)
Dy. HOD - Clinical Chemistry -
NRL



Dr Nimmi Kansal
MD, Biochemistry (DMC - 9550)
Technical Director - Clinical
Chemistry & Biochemical Genetics
NRL - Dr Lal PathLabs Ltd



Dr.Richa Sirohi
MD, Biochemistry (MCI - 15-19066)
Senior Consultant Biochemist
NRL - Dr Lal PathLabs Ltd

-----End of report -----



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IMPORTANT INSTRUCTIONS			
•Test results released pertain to the specimen submitted. •All test results are dependent on the quality of the sample received by the Laboratory . •Laboratory investigations are only a tool to facilitate in arriving at a diagnosis and should be clinically correlated by the Referring Physician .•Report delivery may be delayed due to unforeseen circumstances. Inconvenience is regretted .•Certain tests may require further testing at additional cost for derivation of exact value. Kindly submit request within 72 hours post reporting. •Test results may show interlaboratory variations. •The Courts/Forum at Delhi shall have exclusive jurisdiction in all disputes/claims concerning the test(s) & or results of test(s). •Test results are not valid for medico legal purposes. •This is computer generated medical diagnostic report that has been validated by Authorized Medical Practitioner /Doctor. •The report does not need physical signature. (#) Sample drawn from outside source. If Test results are alarming or unexpected, client is advised to contact the Customer Care immediately for possible remedial action. Tel: +91-11-49885050, Fax: - +91-11-2788-2134, E-mail: lalpathlabs@lalpathlabs.com National Reference lab, Delhi, a CAP (7171001) Accredited, ISO 9001:2015 (FS60411) & ISO 27001:2013 (616691) Certified laboratory.			

